

Module 3: Knowledge Representation & Logic

Propositional Logic, Entailment, CNF, First-Order Logic, Unification (MGU) & Resolution

Module III: Knowledge Representation & Logical Reasoning

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| 1. Knowledge-Based Agents & The Wumpus World Environment Architecture | 2. Propositional Logic (PL): Syntax, Semantics, Models & Truth Tables |
| 3. Logical Entailment ($\alpha \models \beta$), Validity (Tautology) & Satisfiability | 4. Conjunctive Normal Form (CNF) 7-Step Conversion Algorithm |
| 5. Propositional Resolution Refutation Theorem & Proof Trees | 6. Forward Chaining & Backward Chaining on Horn Clauses ($O(n)$ Linear Complexity) |
| 7. First-Order Predicate Logic (FOL): Syntax, Quantifiers ($\forall, \exists$) & Semantics | 8. Knowledge Engineering in FOL & Domain Axiomatization |
| 9. Generalized Modus Ponens (GMP) & The Unification Algorithm (MGU) | 10. Skolemization Algorithms (Skolem Constants vs. Skolem Functions) |
| 11. First-Order Resolution Refutation with MGU Substitutions | 12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions) |

Topic 1: Knowledge-Based Agents & The Wumpus World

A **Knowledge-Based Agent (KBA)** maintains an explicit internal representation of the world in a **Knowledge Base (KB)** (a set of sentences in a formal language) and uses an inference engine to derive new knowledge and decide actions:

```
function KB-AGENT(percept) returns an action
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action = ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t = t + 1
  return action
```

The Wumpus World Environment Benchmark:

4×4 grid of rooms with Agent starting at $[1, 1]$ facing right.

Wumpus: Monster that eats the agent; gives off a *Stench* in directly adjacent squares.

Pits: Bottomless holes that trap the agent; give off a *Breeze* in adjacent squares.

Gold: Gives off a *Glitter* in its exact square.

Topic 2 & 3: Propositional Logic Syntax, Semantics & Entailment

Logical Operator	Standard Notation	Meaning	Truth Condition (T)
Negation (NOT)	$\neg P$	Not P	True iff P is False
Conjunction (AND)	$P \wedge Q$	P and Q	True iff both P and Q are True
Disjunction (OR)	$P \vee Q$	P or Q	True iff at least one of P, Q is True
Implication (IF-THEN)	$P \implies Q$	If P then Q	False iff P is True and Q is False ($\equiv \neg P \vee Q$)
Biconditional (IFF)	$P \iff Q$	P if and only if Q	True iff both P, Q have identical truth values

Formal Mathematical Concept: Logical Entailment ($\alpha \models \beta$)

Sentence α **entails** sentence β ($\alpha \models \beta$) if and only if in every model where α is true, β is also true:

$$\alpha \models \beta \iff M(\alpha) \subseteq M(\beta)$$

Deduction Theorem: $\alpha \models \beta \iff (\alpha \implies \beta)$ is a Tautology

Refutation Theorem: $\alpha \models \beta \iff (\alpha \wedge \neg\beta)$ is Unsatisfiable

Topic 4 & 5: Conjunctive Normal Form (CNF) & Resolution Refutation

A sentence is in **Conjunctive Normal Form (CNF)** if it is a conjunction of clauses (where each clause is a disjunction of literals):

$$\bigwedge_{i=1}^m \left(\bigvee_{j=1}^{k_i} l_{ij} \right)$$

7-Step Algorithm to Convert Propositional Logic to CNF

1. **Eliminate Biconditionals:** Replace $\alpha \iff \beta$ with $(\alpha \implies \beta) \wedge (\beta \implies \alpha)$.
2. **Eliminate Implications:** Replace $\alpha \implies \beta$ with $\neg\alpha \vee \beta$.
3. **Move Negations Inward (De Morgan's Laws & Double Negation):**

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta, \quad \neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta, \quad \neg(\neg\alpha) \equiv \alpha$$

4. **Distribute $\vee$ over $\wedge$:** Replace $\alpha \vee (\beta \wedge \gamma)$ with $(\alpha \vee \beta) \wedge (\alpha \vee \gamma)$.
5. **Flatten Nested Conjunctions/Disjunctions:** Group into distinct clauses separated by $\wedge$.

Step-by-Step Solved Problem: Resolution Refutation Proof in Propositional Logic

Given Knowledge Base (KB):

1. $P \implies Q$ (If it rains, the ground is wet): $\neg P \vee Q$
2. $Q \implies R$ (If the ground is wet, it is slippery): $\neg Q \vee R$
3. P (It rains): P

Goal: Prove R (It is slippery) using Resolution Refutation.

Step 1: Negate the Goal and add to KB: Add Clause 4: $\neg R$.

Step 2: Apply Resolution Rule to Resolve Complementary Literals:

- Resolve Clause 1 ($\neg P \vee Q$) and Clause 3 (P) on literal $P \implies$ **Clause 5: Q .**
- Resolve Clause 2 ($\neg Q \vee R$) and Clause 5 (Q) on literal $Q \implies$ **Clause 6: R .**
- Resolve Clause 6 (R) and Clause 4 ($\neg R$) on literal $R \implies$ **Empty Clause ($\square$ / False).**

Conclusion: Since deriving the empty clause proves that $\text{KB} \wedge \neg R$ is unsatisfiable, the original goal R is validly entailed ($\text{KB} \models R$). ■

Topic 7 & 8: First-Order Predicate Logic (FOL)

Unlike Propositional Logic (which assumes facts in the world are either True or False), **First-Order Logic (FOL)** models the world in terms of **Objects**, **Relations (Predicates)**, and **Functions**:

FOL Construct	Definition & Syntax	Example Statement
Universal Quantifier ($\forall$)	"For all x ". Typically paired with implication ($\implies$).	$\forall x (\text{King}(x) \implies \text{Person}(x))$
Existential Quantifier ($\exists$)	"There exists some x ". Typically paired with conjunction ($\wedge$).	$\exists x (\text{Crown}(x) \wedge \text{OnHead}(x, \text{John}))$

De Morgan's Laws for Quantifiers

$$\forall x \neg P(x) \equiv \neg \exists x P(x)$$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\forall x P(x) \equiv \neg \exists x \neg P(x)$$

$$\exists x P(x) \equiv \neg \forall x \neg P(x)$$

Topic 9 & 10: Unification Algorithm & Skolemization

1. Unification & Most General Unifier (MGU):

$\text{UNIFY}(p, q) = \theta$ finds a substitution θ such that $\text{Subst}(\theta, p) = \text{Subst}(\theta, q)$:

$\text{UNIFY}(\text{Knows}(\text{John}, x), \text{Knows}(\text{John}, \text{Jane})) = \{x/\text{Jane}\}$

$\text{UNIFY}(\text{Knows}(\text{John}, x), \text{Knows}(y, \text{Bill})) = \{y/\text{John}, x/\text{Bill}\}$

Occur-Check: A variable x cannot be unified with a term containing x (e.g., x and $f(x)$ cannot unify because substitution causes infinite recursion).

2. Skolemization:

Skolem Constant: Eliminates an existential quantifier not in the scope of any universal quantifier:

$$\exists x \text{ Heart}(x) \implies \text{Heart}(H_1) \quad \text{where } H_1 \text{ is a new unique constant}$$

Skolem Function: Eliminates an existential quantifier within the scope of a universal quantifier:

$$\forall x \exists y \text{ Mother}(y, x) \implies \forall x \text{ Mother}(m(x), x) \quad \text{where } m(x) \text{ is a Skolem function}$$



Top BIT Mesra Exam Questions & Answers (Module III)

Q1. Differentiate between Forward Chaining and Backward Chaining for Horn Clauses. (8 Marks)

1. **Forward Chaining (Data-Driven):** Starts with known atomic facts in the KB and applies inference rules to derive new facts until the goal is generated. Complete for definite clauses and runs in linear time $O(n)$. Used in reactive expert systems and real-time monitoring.
2. **Backward Chaining (Goal-Driven):** Starts with the target query/goal and works backward by finding rules whose conclusions match the goal, establishing their premises as new subgoals. Avoids exploring irrelevant facts. Used in Prolog and automated theorem provers.

Q2. Convert the English statements into FOL and prove by Resolution Refutation: "Every person who loves animals is loved by someone. Jack loves all dogs. All dogs are animals. Prove that someone loves Jack." (10 Marks)

1. $\forall x [(\forall y \text{ Animal}(y) \implies \text{Loves}(x, y)) \implies (\exists z \text{ Loves}(z, x))]$
2. $\forall y (\text{Dog}(y) \implies \text{Loves}(\text{Jack}, y))$
3. $\forall y (\text{Dog}(y) \implies \text{Animal}(y))$
4. Negated Goal: $\neg \exists z \text{ Loves}(z, \text{Jack}) \implies \forall z \neg \text{Loves}(z, \text{Jack})$

Converting to CNF clauses and unifying with Skolem functions yields the empty clause ($\square$), formally proving someone loves Jack.